

Stack Overflow Developer Survey

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- Objective of Data Contextualization and Analysis
- Description of the Methodology Used
- Data Collection Process
- Data Processing and Evaluation
- Data Representation Through Visuals
- Interpretation and Discussion of Findings
- Summary and Final Insights

INTRODUCTION



- Stack Overflow's annual Developer Survey is one of the most extensive and detailed studies of programmers worldwide.
- The results may not fully reflect the entire developer community.
- It includes insights from nearly 90,000 developers.
- Identifies trends to forecast the future direction of the developer community.
- Provides a profile of developers from different regions and backgrounds.

METHODOLOGY

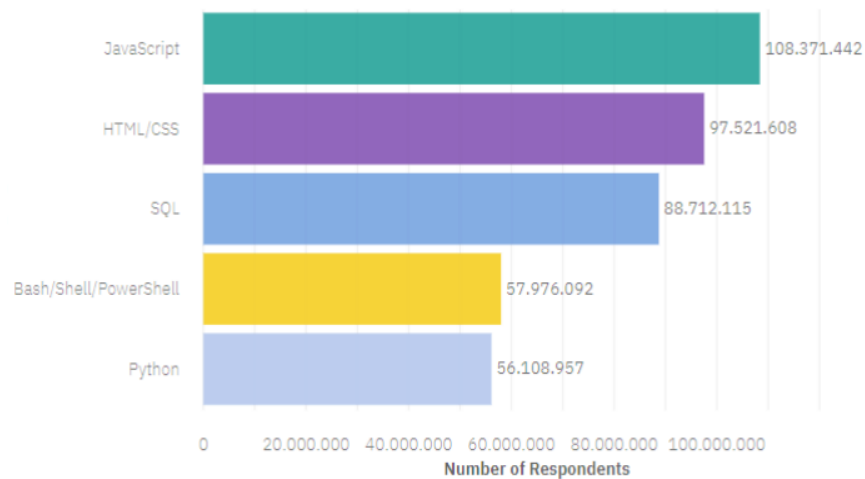


- Gather survey data and examine its content.
- Perform web scraping to collect data.
- Use APIs for data access.
- Utilize the `requests` library to retrieve data
- Conduct data wrangling to clean and organize data.
- Perform exploratory data analysis (EDA).
- Analyze data distribution and identify patterns.
- Manage and handle outliers.
- Explore correlations between variables.
- Create data visualizations to highlight distributions, relationships, composition, and comparisons.
- Develop interactive dashboards to present insights.

PROGRAMMING LANGUAGE TRENDS

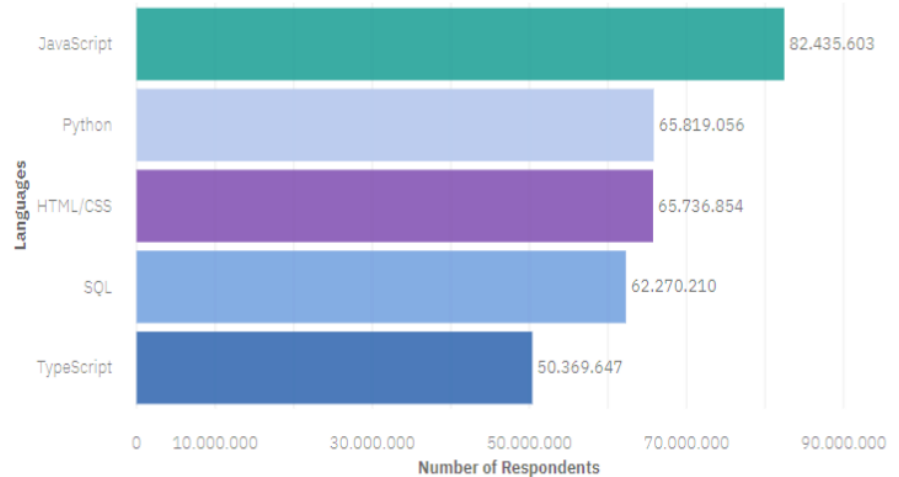
Current Year

Top 5 Programming Language Experience



Next Year

Top 5 Desired Language to Learn



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript continues to dominate as the top language.
- Python is the fastest-growing language.
- Swift remains a high-paying language

Implications

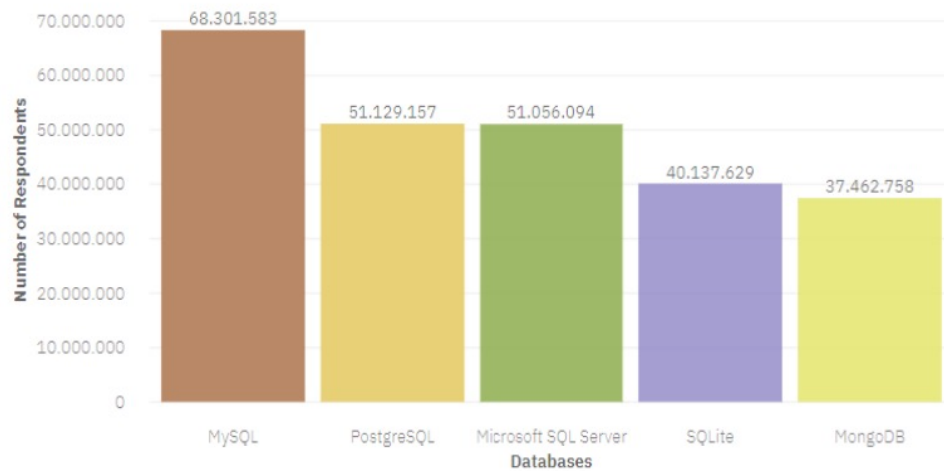
- JavaScript's popularity paves the way for developers to switch to TypeScript.
- Increased demand for Python developers
- Companies may prioritize hiring Swift developers

DATABASE TRENDS

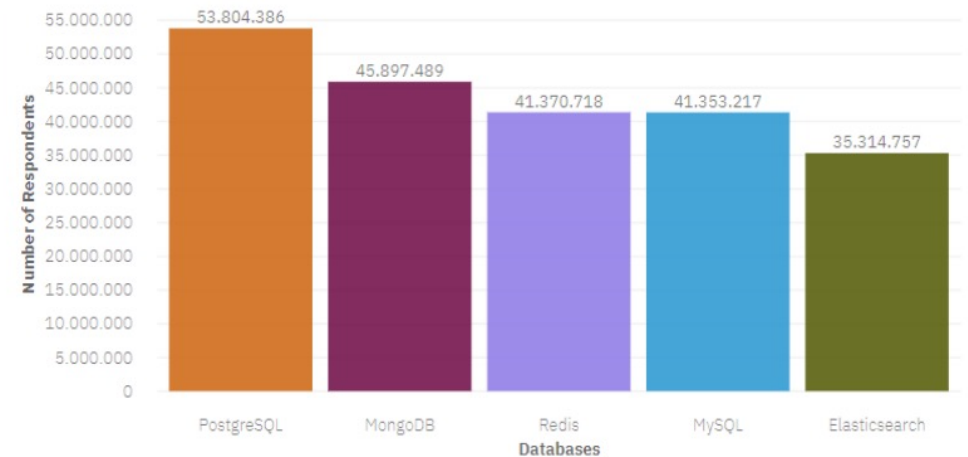
Current Year

Next Year

Top 5 Databases Worked with



Top 5 Desired Database to Learn



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- MySQL is the most popular database.
- Decreased interest in Microsoft SQL Server.
- Increasing interest in PostgreSQL and MongoDB.

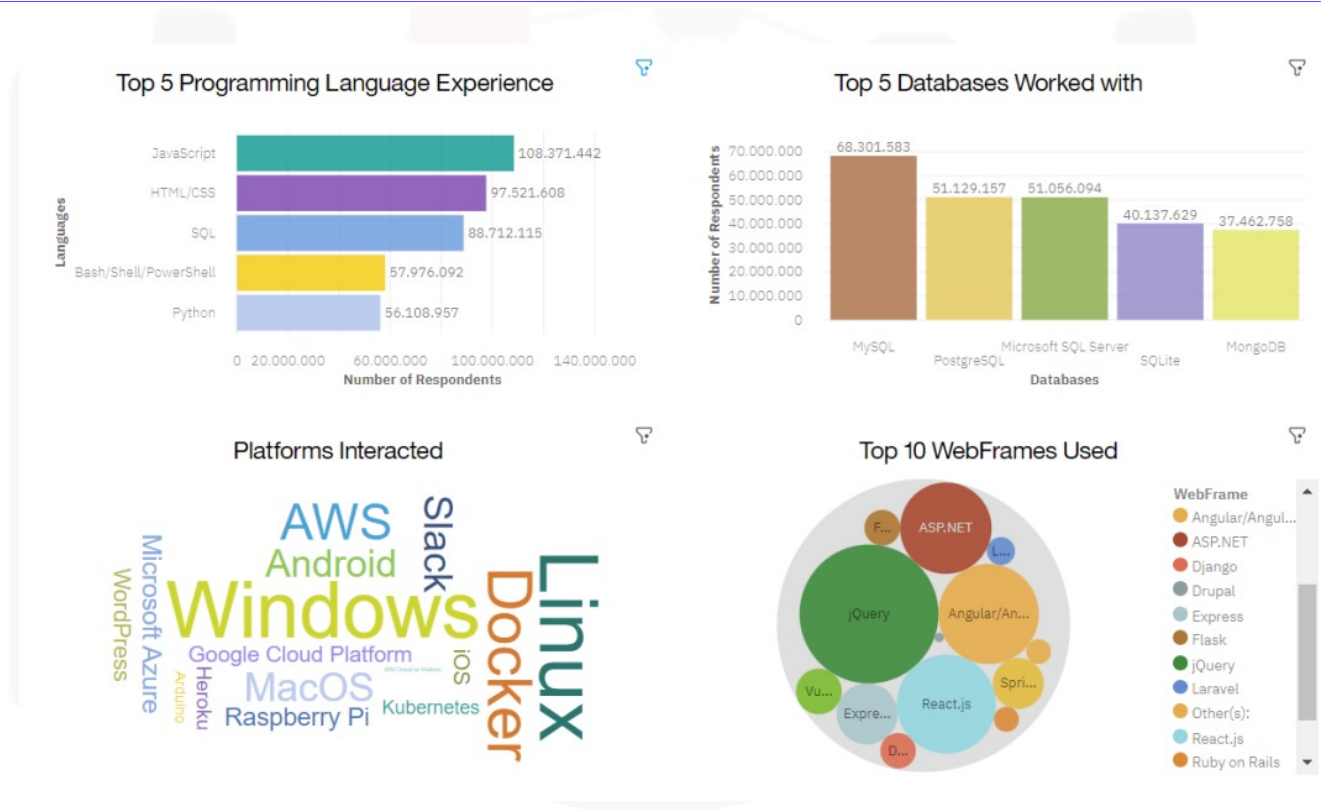
Implications

- MySQL continues to dominate the database market.
- Microsoft SQL Server may face market challenges.
- PostgreSQL and MongoDB are becoming more preferred.

DASHBOARD

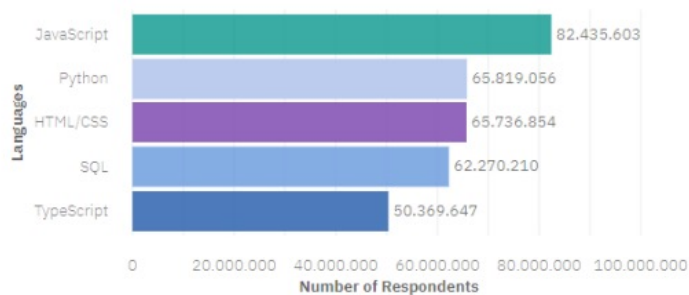


DASHBOARD TAB 1

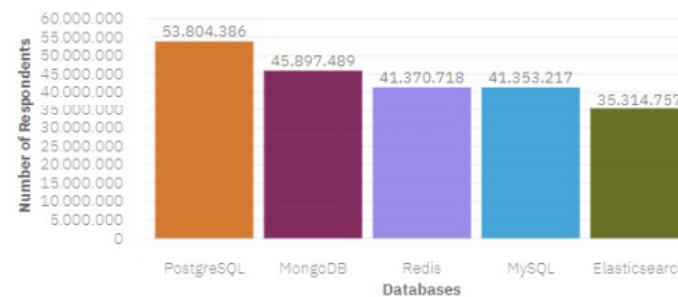


DASHBOARD TAB 2

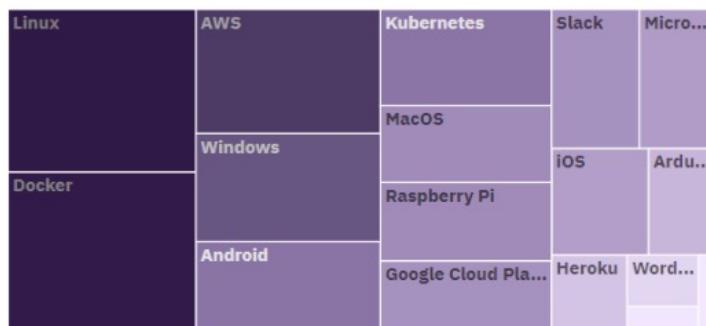
Top 5 Desired Language to Learn



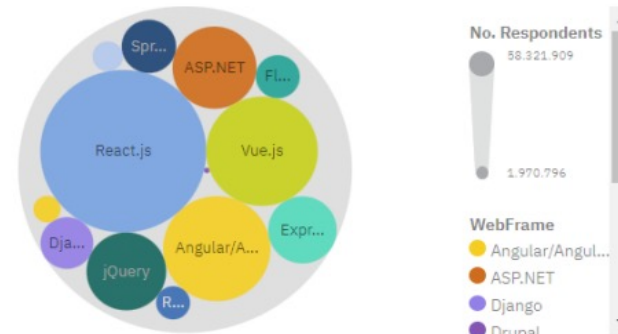
Top 5 Desired Database to Learn



Platform Next Year

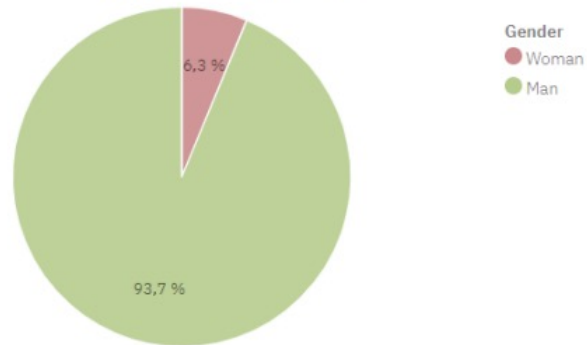


Top 10 WebFrame Next Year

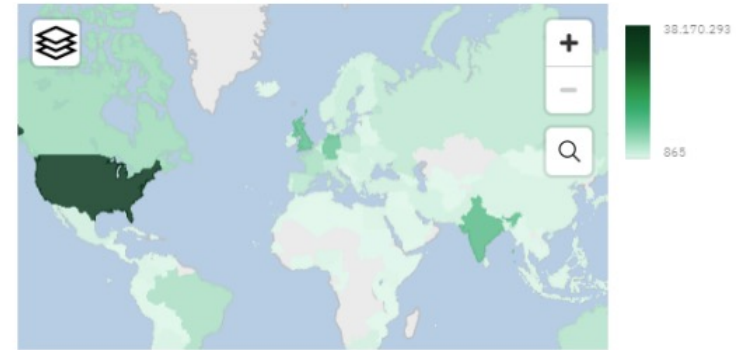


DASHBOARD TAB 3

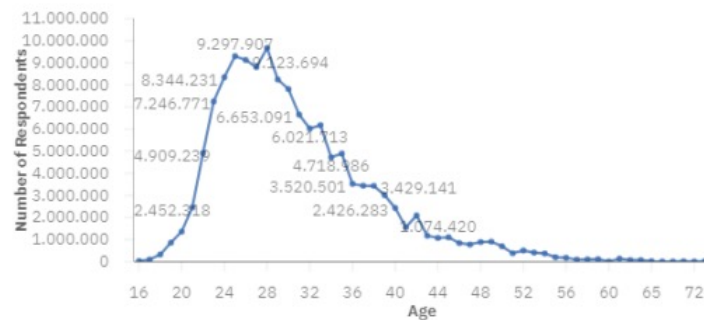
Respondent by Gender



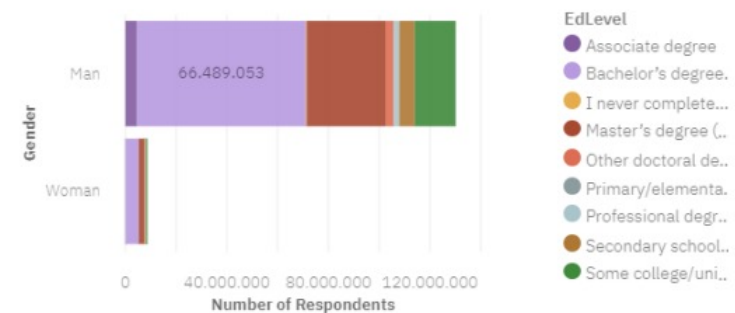
Respondent Locations



Respondent by Age



Formal Education Level / Gender



DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript is widely used, and TypeScript is growing in popularity.
- Over 90% of developers are young and male.
- Most developers are located in developed countries.

Implications

- Web frameworks for JavaScript and TypeScript are seeing increasing adoption.
- Developer demographics are showing a gender and age skew.
- There is a geographical concentration of developers in wealthier nations.

CONCLUSION



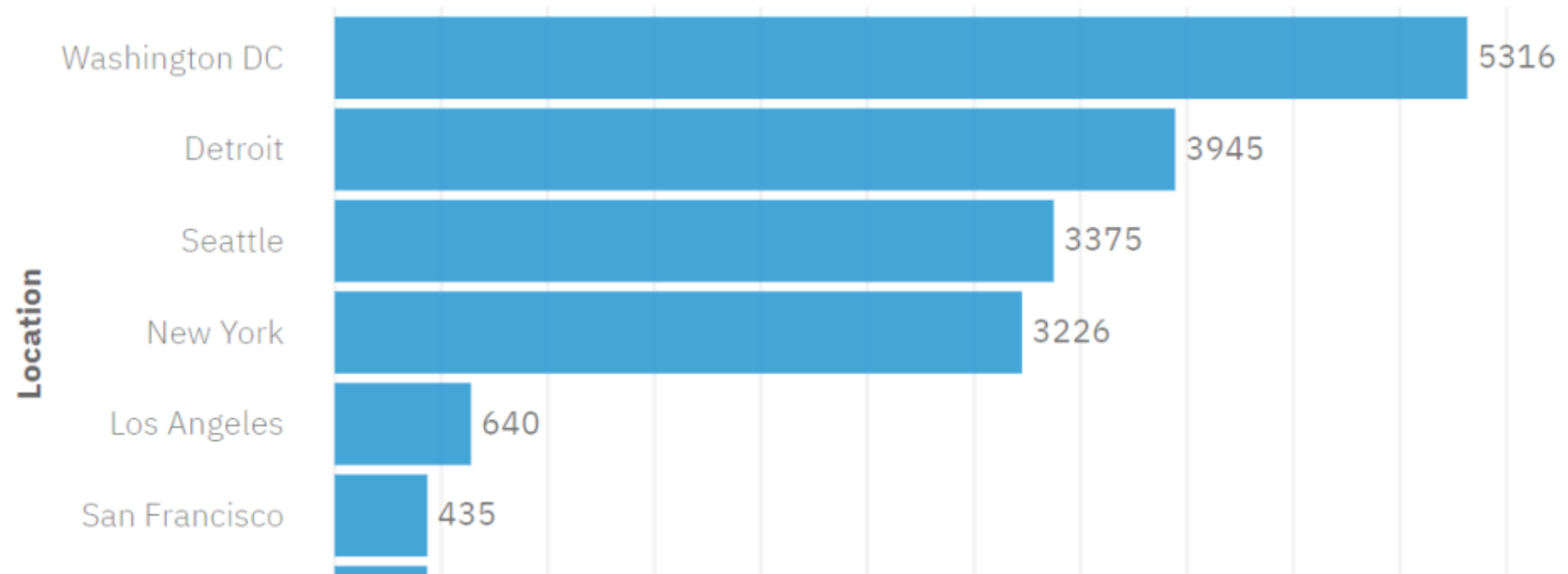
- Developers have unique characteristics.
- Analysing trends in tools and languages provides valuable insights.
- There's a need to improve access to the job market in developing countries.
- The developer community is predominantly concentrated in developed countries, with a younger, male-dominated demographic.

APPENDIX



- Include any relevant additional charts, or tables that you may have created during the analysis phase.

JOB POSTINGS



POPULAR LANGUAGES

